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Letter of reference for Dr Alexis Gomel

Alexis Gomel joined my research group in May 2018 and defended his PhD Thesis in March 2023.

He began the PhD program by attending the advanced school *Wave Turbulence and Extreme Events* in June 2018. Since Alexis had already developed experience in dynamical systems during his Master's thesis, the initial phase of his PhD project focused on testing various early warning indicators near bifurcations—particularly kurtosis and relative tail weights—using toy models and data from an optical experiment. The aim was to characterize the parameter region where multiple filamentations occur under the influence of turbulence in high-intensity laser pulses, and to detect transitions between filamenting and non-filamenting regimes through early warning indicators. A paper on this topic was later published. This was the first occasion where I could observe his working style: *he organized the experimental procedures with great precision, paid close attention to all possible sources of error, and consistently proposed improvements to better characterize and isolate the processes under study.* He also investigated the behavior of several toy models featuring deterministic bifurcations, noise, and time-dependent parameters, in order to test early warning indicators under general conditions, including noise-induced and rate-induced tipping mechanisms. *He developed Python codes for these toy models using packages for solving stochastic differential equations.*

After this first phase, we started a collaboration with Prof. Amin Chabchoub at the University of Sydney, aimed at conducting experiments using the wave flume facility there. *Alexis actively participated in all phases of the project, from the initial discussions on the theoretical framework to the design and installation of measurement devices in the flume.* Building on the analogy between optical filamentation and the propagation of surface gravity waves in a water tank—both of which are, at leading order, described by the nonlinear Schrödinger equation (NLS)—we decided to investigate how the nonlinear behavior of waves is affected by a slowly varying external parameter: the bottom slope of the tank.

In November 2019, Alexis spent one month in Sydney to carry out the experiments with the support of Prof. Chabchoub. As with the previous optical experiments, he was extremely meticulous in documenting all the different configurations to be tested. Prior to his departure, he prepared MATLAB scripts to facilitate comparisons between experimental setups and theoretical initial conditions, as well as to analyze the experimental outputs. This preparation allowed him to adapt in real time: correcting and rerunning experiments, introducing additional tests by varying the initial wave steepness, frequency, or the slope of the tank bed, and estimating the dissipation and main physical constraints involved. *By the end of the campaign, he had gathered a large and comprehensive dataset. The pandemic forced him to analyze the data independently, and he did so with great autonomy.* As predicted by the NLS theory, the change in bathymetry induced a change in the wave evolution: at the

point of maximum amplification, the wave packet ‘froze’ into a new solution with constant amplitude. This result provided experimental evidence of the possibility to manipulate the phase space and induce a transition toward a stable solution with specific characteristics through controlled variations in bathymetry. *The success of this experimental campaign was largely due to the rigorous methodology developed by Alexis: minimizing losses, reducing errors, optimizing the setup, analyzing data, developing MATLAB tools, and comparing the results with numerical simulations.* His work culminated in high-quality publications in *Physical Review Letters* and *Nonlinear Dynamics* in the following years.

In the final year of his PhD, Alexis continued working with the data collected in Sydney. He identified that previous theoretical models did not properly account for higher-order terms within the NLS framework, particularly in intermediate and deep water conditions. *A theoretical paper was subsequently published, demonstrating the importance of accurately incorporating the effect of the mean flow.* This work was carried out in collaboration with Corentin Montessuit, a Master’s student in Physics, *whose research Alexis closely supervised and supported*, especially during the challenging period of the pandemic.

Over the five years of his PhD, Alexis significantly enhanced his skills in data analysis using both MATLAB and Python, as well as deepened his understanding of dynamical systems theory. He also served for one year as *teaching assistant* for the exercise sessions of the *Nonlinear Systems* course at the Master level, taught by myself and Prof. Jérôme Kasparian in the Physics Section. He published his results in open-access, peer-reviewed journals and presented his work at major international conferences, including EGU 2022 (online) and the 10th European Nonlinear Dynamics Conference (ENOC 2022).

Alexis has an open and collaborative attitude, and he is passionate about both teaching and working in a team.

Sincerely,

Maura Brunetti

